
Cybersecurity Training Diary

Day 2: CIA Triad (Confidentiality, Integrity, Availability)

1. Topics Covered

✓ What is the CIA Triad?

- The foundational model of cybersecurity: **Confidentiality, Integrity, Availability**.

✓ Deep Dive into Each Principle:

- **Confidentiality:**
 - Ensures data is accessible **only to authorized users**.
 - Tools: Encryption (AES, RSA), Access Controls (Passwords, MFA).
 - Example: Encrypted messaging (WhatsApp, Signal).
- **Integrity:**
 - Ensures data is **accurate and unaltered** during storage/transmission.
 - Tools: Hash functions (SHA-256), Digital Signatures.
 - Example: Detecting file tampering via checksums.
- **Availability:**
 - Ensures systems/data are **accessible when needed**.
 - Threats: DDoS attacks, hardware failures.
 - Solutions: Redundancy, backups, disaster recovery plans.

✓ Real-World Violations:

- **Confidentiality Breach:** Facebook-Cambridge Analytica data leak.
- **Integrity Breach:** Stuxnet malware altering industrial systems.
- **Availability Breach:** 2021 AWS outage disrupting major websites.

2. Key Takeaways

No security without CIA: All three principles must be balanced.

Trade-offs Exist: Strong encryption (Confidentiality) may slow systems (Availability).

Beyond IT: CIA applies to physical security (e.g., locked server rooms).

3. Practical Exercise

Task 1: Use `md5sum` (Linux) or `CertUtil -hashfile` (Windows) to verify file integrity.

```
md5sum important_document.txt # Compare hashes before/after transfer.
```

Task 2: Simulate a DDoS attack (ethically!) using **LOIC** (in a lab environment).

4. Questions for Reflection

Can you think of a system where **Availability** is more critical than Confidentiality? (Hint: Emergency services)

How would you explain the CIA Triad to a non-technical executive?

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